Exact Antisymmetric-Singlet Ground States and Bell Correlations in Finite-Hilbert-Space Substrate Models

A Conservative Framework for Emergent Quantum Structure

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Abstract

We investigate how quantum correlations and particle-like structure emerge in finite-dimensional quantum systems governed by local Hamiltonians and constraint operators. Working entirely within orthodox quantum mechanics, we study two complementary phenomena: (i) Bell-CHSH correlations in gauge-constrained substrates, and (ii) exact antisymmetric exchange symmetry in two-excitation systems.

The substrate framework employs three conservative axioms: finite quantum substrate, unitary evolution, and emergent structure. No assumptions are made about the Standard Model, spacetime, or quantum gravity. All claims concern only well-specified toy models that can be simulated exactly.

Key results on Bell correlations: Constraint-driven dynamics generate CHSH violations up to the Tsirelson bound ($|S| = 2\sqrt{2}$) across wide parameter ranges. Both generic perturbations and explicit information extraction degrade Bell correlations in correlation with entropy production.

Key results on fermionic-like structure: Two-excitation Hamiltonians with defrag attraction, Gauss constraints, and antiferromagnetic spin coupling produce ground states exhibiting: (i) exact antisymmetric exchange symmetry ($w_A = 1.000$ to machine precision), (ii) pure spin-singlet character ($F_s = 1.000$), and (iii) Tsirelson-saturating Bell correlations ($|S| = 2.8284...$). Multi-parameter robustness testing across a six-dimensional hypercube with $\pm 20\%$ perturbations reveals these properties persist across all 100 sampled Hamiltonians, while ground state energies vary significantly.

Methodology validation: We document the discovery and correction of two diagnostic bugs during investigation, demonstrating that reported results survive rigorous validation including deliberate symmetry-breaking tests. When spin interactions were flipped from antiferromagnetic to ferromagnetic, diagnostics correctly showed collapse of singlet character and Bell violations, confirming measurement reliability.

A complementary three-quark "proton" model demonstrates that the same constraint language supports bound-state-like structure with centrally-peaked density profiles and nontrivial spin statistics.

Guiding principle: Throughout, we maintain a conservative posture—*if the data do not show it, we do not say it.*

Contents

1 Introduction

Modern quantum theory is defined as much by the *constraints* it imposes as by the dynamical laws it supplies. The standard formalism combines structural requirements—linear Hilbert space, unitary evolution, Born rule—with empirical constraints such as no-signalling and the Tsirelson bound on Bell inequality violations. These carve out a narrow region of correlation space: classical hidden-variable models lie below the Bell bounds, while generic nonlocal theories exceed the quantum limit.

The substrate framework studied here takes a deliberately modest approach. Instead of postulating new primitives beyond standard quantum theory, it starts from finite-dimensional Hilbert spaces, local Hamiltonians, and explicit constraint operators, and asks: *What kinds of effective behavior can emerge?*

1.1 Scope and Objectives

This work has three primary objectives:

1. **Bell correlations**: Study how constraint-driven substrate dynamics balance correlation, energy, and information flow, using CHSH as a diagnostic rather than a binary locality test.

2. **Fermionic-like structure**: Investigate whether exact antisymmetric exchange symmetry and exclusion-like behavior can emerge from two-body Hamiltonians without imposing fermionic statistics by hand.

3. **Methodological rigor**: Demonstrate transparent, reproducible computational physics including systematic bug discovery, validation through symmetry breaking, and conservative interpretation.

1.2 Organization and Evidence Hierarchy

The program is organized into three tiers:

- **Tier I**: Non-speculative mathematical definitions

- **Tier II**: Numerical or structural observations
- **Tier III**: Explicitly flagged interpretive hypotheses

Only Tier II is used as evidence. Tier III is presented as conjecture to guide future work. **Guiding principle**: *If the data do not show it, we do not say it.*

1.3 Central Operational Questions

For Bell correlations: Can a purely constraint-driven substrate generate Bell violations that saturate the Tsirelson limit and degrade in a controlled way under information extraction?

For fermionic structure: Can two-body Hamiltonians produce ground states with exact antisymmetric exchange symmetry and singlet pairing without postulating fermionic statistics at the fundamental level?

The simulations presented below answer both questions in the affirmative for specific families of finite models.

2 Framework

2.1 Axioms and Scope

The substrate program is defined by three conservative axioms:

Axiom 1: Finite Quantum Substrate

The fundamental degrees of freedom live in a finite-dimensional Hilbert space $\mathcal{H}$ built from local factors. There is no continuum limit assumed at the outset and no additional ontic primitives beyond the quantum state.

Axiom 2: Unitary Evolution

Dynamics are generated by a time-independent Hermitian Hamiltonian $\hat{H}$ that acts locally on the Hilbert space. Evolution is given by the Schrödinger equation with $U(t) = e^{-i\hat{H}t}$.

Axiom 3: Emergent Structure

Classical-looking variables, effective fields, or "objects"—including anything that might be called a particle or pointer—are interpreted as coarse-grained, dynamically stable patterns in the underlying quantum state. There is no additional notion of collapse or classicality beyond this emergent description.

These axioms stay strictly within orthodox quantum mechanics. The question is: *How far can we go with only these ingredients, plus specific choices of Hamiltonian and constraint operators?*

2.2 Finite-Hilbert Substrates

Concrete models live on finite lattices:

$$\mathcal{H} = \mathcal{H}_{\text{sites}} \otimes \mathcal{H}_{\text{links}} \otimes \cdots \tag{1}$$

where site factors describe local matter degrees of freedom and link factors encode $\mathbb{Z}_2$-like gauge or auxiliary variables.

For Bell experiments: 2×2 spatial lattices with modest numbers of matter and link qubits.

For fermionic emergence: $2 \times 2 \times 2$ lattices in the two-excitation sector.

For bound states: $3 \times 3 \times 3$ lattices with matrix-free Hamiltonian representations.

All structure is encoded in the Hamiltonian, which is a sum of terms acting nontrivially only on small regions.

2.3 Constraints and Locality

2.3.1 Local Gauss-like Constraints

At each lattice site s, define:

$$\hat{G}_s = (-1)^{n_s} \prod_{\ell \in \mathrm{star}(s)} \sigma^x(\ell) \tag{2}$$

where n_s is the occupation number and the product runs over incident links. This structure is directly inspired by $\mathbb{Z}_2$ Gauss laws in lattice gauge theory.

Violations are penalized energetically:

$$\hat{H}_{\mathrm{Gauss}} = \lambda_G \sum_s (I - \hat{G}_s)^2 \tag{3}$$

For $\lambda_G \gg 0$, inconsistent matter-link patterns become very costly. For $\lambda_G = 0$, the constraint is inactive.

2.3.2 Information Routing Interpretation

Because $\hat{G}_s$ couples matter occupancy to nearby links, it can be viewed as a rule governing how information about local configuration is routed through neighboring degrees of freedom. This interpretation is used purely as a structural lens; no claim is made that it reflects a fundamental law of nature.

3 Two-Body Fermionic Emergence

3.1 Motivation and Context

The standard framework of quantum field theory treats fermionic statistics as fundamental, with the spin-statistics theorem connecting half-integer spin to antisymmetric wavefunctions. While rigorous within relativistic QFT, this leaves open: *Could fermionic behavior emerge from more primitive quantum structures?*

We investigate two-excitation Hamiltonians on finite 3D lattices. These models are built from the three axioms and do **not** postulate fermionic statistics or impose antisymmetrization by hand.

3.2 Hilbert Space Structure

Two-excitation sector on an $L_x \times L_y \times L_z$ cubic lattice. Basis states:

$$|\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2\rangle \tag{4}$$

where $\mathbf{r}_i$ denotes lattice site and $\sigma_i \in \{\uparrow, \downarrow\}$ is spin-1/2. Full Hilbert space dimension: $(L_x L_y L_z)^2 \times 4$.

3.3 Hamiltonian Structure

Five terms:

$$\hat{H} = \hat{T} + \hat{M} + \hat{V}_{\text{defrag}} + \hat{V}_{\text{Gauss}} + \hat{V}_{\text{spin}} \tag{5}$$

Kinetic term: Nearest-neighbor hopping, $J_{\text{hop}} = 1.0$
Mass term: Onsite energy, $m = 0.1$
Defrag interaction: Nonlocal density attraction via Gaussian kernel:

- Strength: $g_{\text{defrag}} = 1.0$ (attractive)
- Width: $\sigma_{\text{defrag}} = 1.0$

Gauss constraint: Local density-squared penalty, $\lambda_G = 5.0$
Spin interactions: SU(2)-invariant terms:

- Onsite coupling: $\lambda_S = -1.0$ (antiferromagnetic, singlet-favoring)
- Exchange coupling: $J_{\text{exch}} = 1.0$

Baseline system: $L_x = L_y = L_z = 2$. Ground states via exact diagonalization.

3.4 Fermionic Diagnostics

3.4.1 Exchange Antisymmetry

Particle exchange operator swaps labels 1 and 2. Projectors:

$$\hat{P}_A = \frac{1}{2}(I - \hat{P}_{\text{swap}}) \tag{6}$$

$$\hat{P}_S = \frac{1}{2}(I + \hat{P}_{\text{swap}}) \tag{7}$$

Antisymmetry weight for ground state $|\Psi_0\rangle$:

$$w_A = \langle \Psi_0 | \hat{P}_A | \Psi_0 \rangle \in [0, 1] \tag{8}$$

with $w_A = 1$ meaning perfect antisymmetry, $w_A + w_S = 1$.

3.4.2 Spin Singlet Character

SU(2) basis:

$$|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad \text{(singlet)} \tag{9}$$

$$|T_{+1}\rangle = |\uparrow\uparrow\rangle \tag{10}$$

$$|T_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \tag{11}$$

$$|T_{-1}\rangle = |\downarrow\downarrow\rangle \tag{12}$$

Singlet fraction F_s measures weight in $|S\rangle$ at overlap sites.

Critical distinction: Naive heuristics classifying opposite spins as "singlet" cannot distinguish true singlet from triplet $m = 0$ state. Proper SU(2) projectors are required.

3.4.3 Bell-CHSH Correlations

Trace out spatial degrees to get reduced two-spin density matrix. CHSH parameter:

$$S = E(\mathbf{a}, \mathbf{b}) + E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) - E(\mathbf{a}', \mathbf{b}') \tag{13}$$

Tsirelson bound: $|S|_{\max} = 2\sqrt{2} \approx 2.828$ (maximally entangled states).

3.5 Baseline Results

For the parameter regime studied:

Baseline Ground State Properties

- Antisymmetry weight: $w_A = 1.000000$ (to machine precision $\sim 10^{-15}$)
- Singlet fraction: $F_s = 1.000000$
- Bell-CHSH parameter: $|S| = 2.8284271...$ (saturates Tsirelson bound)

The ground state lies *exactly* in the antisymmetric-singlet sector and exhibits maximal quantum entanglement.

4 Multi-Parameter Robustness Testing

4.1 Initial Suspicion

The perfect uniformity of baseline results raised immediate concern. True physical robustness should show *some* variation. This prompted systematic validation following the principle: **If results look too good to be true, they probably are (bugs or artifacts)**.

4.2 Robustness Protocol

Six-dimensional parameter space:

$$\boldsymbol{\theta} = (g_{\text{defrag}}, \sigma_{\text{defrag}}, \lambda_G, \lambda_S, \lambda_T, J_{\text{exch}}) \tag{14}$$

Perturbation scales: $\delta \in \{0.10, 0.20\}$
Each sample:

$$\theta_i^{(\text{sample})} = \theta_i^{(\text{baseline})}(1 + \epsilon_i), \quad \epsilon_i \sim \text{Uniform}(-\delta, \delta) \tag{15}$$

Euclidean distance: $\|\boldsymbol{\epsilon}\|_2 \approx \sqrt{6}\delta$

- $\delta = 0.10$ corresponds to $\sim 25\%$ L^2 distance
- $\delta = 0.20$ corresponds to $\sim 50\%$ L^2 distance

4.3 Pass Criteria

A sample passes if:

- $w_A > 0.95$ (antisymmetric)
- $F_s > 0.95$ (singlet-dominant)
- $|S| > 2.0$ (Bell-violating)

4.4 Robustness Results

Metric	$\delta = 0.10$	$\delta = 0.20$
Fraction $w_A > 0.95$	1.000	1.000
Fraction $F_s > 0.95$	1.000	1.000
Fraction $\lvert S\rvert > 2.0$	1.000	1.000
Mean w_A	1.000	1.000
Mean F_s	1.000	1.000
Mean $\lvert S\rvert$	2.8284	2.8284
Std(E_0)	0.256	0.512

Table 1: Robustness scan results across 100 samples at two perturbation scales. All samples exhibit exact fermionic quantum numbers to machine precision, while ground state energies vary significantly.

Critical observation: Ground state energies *do* vary substantially (standard deviations $\sim 3-6\%$ of baseline), demonstrating that different Hamiltonians produce genuinely distinct quantum states. Yet fermionic quantum numbers remain exact.

5 Diagnostic Validation and Bug Discovery

5.1 Methodology: Suspicious of Perfection

Perfect uniformity across 200 samples suggested either:

1. Genuine structural stability (symmetry protection)
2. Diagnostic bugs masking real variation

Following conservative principles, we assumed bugs until proven otherwise.

5.2 Symmetry-Breaking Validation Test

To verify diagnostic responsiveness, we deliberately broke symmetries:

$$\lambda_S = +1.0 \text{ (ferromagnetic)}, \quad J_{\text{exch}} = 0.0 \tag{16}$$

Expected behavior: Diagnostics should show departure from perfect antisymmetry/singlet/Tsirelson values.

5.3 Bug #1: Exchange Antisymmetry Normalization

Symptom: Symmetry-breaking test produced:

$$w_A = -3.000, \quad w_S = 1.000 \tag{17}$$

These values cannot represent projector weights ($w_A, w_S \in [0, 1]$ by definition).

Root cause: Code computed raw squared norms but printed them without normalization. Since the sum equals 4, raw values lived in $[0, 4]$ rather than $[0, 1]$.

Fix: Proper normalization ensures $w_A, w_S \in [0, 1]$ and $w_A + w_S = 1$.

5.4 Bug #2: Singlet/Triplet Decomposition

Symptom: After fixing Bug #1, symmetry-breaking test showed $w_A \approx 0.22$ (correct) but still reported $F_s = 1.000$ (impossible for ferromagnetic regime).

Root cause: Initial implementation classified opposite spins as "singlet" and same spins as "triplet." This naive heuristic cannot distinguish:

- True singlet: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$
- Triplet $m = 0$: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$

Fix: Implement proper SU(2) projectors for all four spin states.

5.5 Post-Fix Validation

After corrections, symmetry-breaking test produced:

$$w_A \approx 0.22, \quad F_s \approx 0.06, \quad |S| \approx 2.28 \tag{18}$$

All diagnostics now respond appropriately to symmetry breaking.

Critically, re-running full robustness scans with corrected diagnostics reproduced original results: all samples remain at $w_A = 1.0$, $F_s = 1.0$, $|S| = 2.8284$ to machine precision.

Key Methodological Lesson

The "too perfect" results were real physics measured by buggy diagnostics. Fixing bugs and seeing the same results was the ultimate validation—not despite the bugs, but *because* finding and fixing them proved the robustness of underlying physics.

6 Physical Interpretation

6.1 Origin of Structural Stability

The persistence of exact fermionic quantum numbers suggests ground states lie in a symmetry-protected subspace. Hamiltonian structure contains:

1. **SU(2) spin symmetry**: All spin terms respect $[\hat{H}, \hat{\mathbf{S}}^2] = 0$, forcing eigenstates into definite spin irreps.
2. **Exchange structure**: Combination of defrag attraction, Gauss repulsion, and exchange coupling energetically favors antisymmetric sector for ground state.
3. **Attractive nonlocality**: Defrag term creates effective binding while Gauss constraint prevents collapse.

For the parameter regime studied ($\lambda_S < 0$, $J_{\text{exch}} > 0$), these combine to lock ground state into antisymmetric-singlet sector. Perturbations preserving relative signs and rough magnitudes maintain this structure.

6.2 What We Can Claim

Validated Claims

- Exact antisymmetric exchange symmetry in ground states
- Pure spin-singlet character and Tsirelson saturation
- Robustness across $\sim 50\%$ L^2 perturbations in 6D parameter space
- Diagnostic responsiveness to deliberate symmetry breaking
- Distinct quantum states with identical fermionic quantum numbers

6.3 What We Cannot Claim

Beyond Current Scope

- Connection to relativistic fermions (model is nonrelativistic)
- Full spin-statistics theorem (we work outside its assumptions)
- Anyonic or fractional statistics (two-body sector only)
- Scalability to many-body systems
- "Universality class" without analytical proof
- "Topological protection" without formal demonstration

6.4 What Remains Unknown

- Analytical proof of symmetry protection mechanism
- Precise boundaries of robust parameter region
- Behavior under disorder or symmetry-breaking perturbations
- Connection to topological invariants, if any
- Extension to three-body and many-body systems

7 Bell Correlations: Constraint-Driven Dynamics

7.1 Bell Experiment Mapping

Two spatially separated "measurement regions" A and B on lattice. Each region supports two measurement settings: A_0, A_1 on region A and B_0, B_1 on region B.

Prepare initial state $|\psi_0\rangle$, evolve under substrate Hamiltonian for time T, evaluate:

$$E_{ij} = \langle \psi(T) | A_i B_j | \psi(T) \rangle, \quad i, j \in \{0, 1\} \tag{19}$$

CHSH parameter:

$$S = E_{00} + E_{01} + E_{10} - E_{11} \tag{20}$$

Classical bound: $|S| \leq 2$. Tsirelson bound: $|S| \leq 2\sqrt{2}$.

7.2 Gauss Sweep Results

As λ_G increases from zero, $|S(\lambda_G)|$ exhibits three regimes:

- $\lambda_G = 0$: Gauss penalty inactive, yet $|S| > 2$ (Bell violation present)
- $\lambda_G \approx 3$: Maximum $|S|_{\max} \approx 2.8284$ (Tsirelson saturation)
- $\lambda_G \gg 3$: System frozen into low-violation subspaces, $|S|$ decreases

Key finding: Broad plateau of λ_G values supporting robust Tsirelson-level correlations.

7.3 Exchange Sweep Results

Remarkably, model exhibits Bell violation even when exchange coupling is turned off:

$$|S(J_{\text{exch}} = 0)| \approx 2.8 > 2 \tag{21}$$

Maximum $|S|_{\max} \approx 2.8284$ near $J_{\text{exch}} \approx 0.3$.

Key observation: Finite exchange channel not strictly required. Constraint structure alone generates nonclassical behavior; exchange term fine-tunes within quantum-allowed region.

7.4 Sequential Perturbation Test

Start from state with $|S| \approx 2.8$, apply series of small perturbations. For each step k:

- Record entropy change ΔS_k^{vN}
- Measure overlap O_k with original state
- Compute change in CHSH $\Delta S_k = S_k - S_0$

Empirical correlations:

- Overlap O_k anti-correlated with $|\Delta S_k^{\text{vN}}|$
- Overlap O_k anti-correlated with $|\Delta S_k|$

Larger perturbations push state further from original configuration in both Hilbert-space overlap and Bell correlations.

7.5 Information Extraction Test

Couple explicit classical readout channel to substrate. For measurement steps k:

- Classical mutual information ΔI_k gained
- Change in substrate entropy ΔS_k^{vN}
- Change in CHSH parameter ΔS_k

Result: Both ΔS^{vN} and $|\Delta S|$ grow monotonically with I_{total}. In small-information regime ($I_{\text{total}} \lesssim 0.3$ bits):

$$\Delta S \approx a I_{\text{total}}, \quad \Delta S^{\text{vN}} \approx a' I_{\text{total}} \tag{22}$$

No free peek: Cannot extract substantial classical information while keeping $|S|$ pinned at Tsirelson bound.

8 Complementary Demonstration: Three-Quark Bound State

To illustrate that the same substrate language supports bound-state-like structure, we construct a three-quark toy model on a finite $3 \times 3 \times 3$ lattice.

8.1 Hamiltonian

$$\hat{H} = \hat{H}_{\text{hop}} + \hat{H}_{\text{mass}} + \hat{H}_{\text{defrag}} + \hat{H}_{\text{Gauss}} + \hat{H}_{\text{Zeeman}} \tag{23}$$

Hilbert space dimension: $(2N_s)^3 = 157,464$ for $N_s = 27$ sites.

Matrix-free implementation: wavefunction stored as tensor $\psi_{r_1 s_1, r_2 s_2, r_3 s_3}$. Hamiltonian-vector products computed via vectorized operations. Ground state via iterative eigensolver.

8.2 Ground State Properties

For representative parameters ($J_{\text{hop}} = 1.0$, $m = 0.1$, $g_{\text{defrag}} = 2.0$, $\lambda_G = 5.0$):

- Ground state energy: $E_0 \approx -11.92$
- Density strongly peaked at central site: $n(1,1,1) \approx 0.23$
- Approximately spherically symmetric profile
- Effective radius: $r_{\text{eff}} \approx 1.77$ lattice units
- Nontrivial spin distribution: $\langle S_z \rangle \approx 0.033$, $\langle S_z^2 \rangle \approx 0.60$

Conclusion: Framework hosts both Bell-type nonclassical correlations and spatially localized composite objects within unified language.

9 Experimentally Testable Prediction

The information-extraction simulations suggest a concrete experimental prediction:

Prediction: Information-Disturbance Tradeoff

Consider any physical CHSH experiment achieving $|S|$ within small tolerance of Tsirelson bound $2\sqrt{2}$. Augment setup with controllable weak measurement channel on one wing. From output, construct classical bit string carrying ΔI bits of mutual information per trial about latent variable.

Then, in regime where added channel does not dominate dynamics:

(1) Heat dissipation: Average heat dissipated per trial increases monotonically with extracted information:

$$\langle \Delta Q \rangle \gtrsim k_B T_{\text{eff}} (\ln 2) \Delta I \tag{24}$$

(2) Bell deficit: Bell parameter decreases monotonically with extracted information. In small-information regime:

$$2\sqrt{2} - |S(\Delta I)| \propto \Delta I \tag{25}$$

No regime exists where $|S| \approx 2\sqrt{2}$ is maintained while ΔI increases for free.

10 Discussion

10.1 Main Results Summary

Fermionic emergence: Two-excitation Hamiltonians with appropriate constraint structure produce ground states with exact antisymmetric exchange symmetry, pure singlet character, and Tsirelson-saturating Bell correlations. These properties are structurally stable across $\sim 50\%$ L^2 perturbations in 6D parameter space.

Bell correlations: Constraint-driven substrate dynamics generate Tsirelson-level CHSH violations across wide parameter ranges. Both perturbations and information extraction degrade correlations in correlation with entropy production.

Bound states: Same constraint language supports localized composite objects with nontrivial spin statistics.

Methodology: Systematic bug discovery and correction via symmetry-breaking validation demonstrates that results survive rigorous scrutiny. Diagnostics respond correctly when physics changes.

10.2 Existence Proof Interpretation

We interpret these results as *existence proofs*:

- Fermionic-like quantum behavior (antisymmetric exchange, singlet pairing, maximal entanglement) *can* emerge from substrate dynamics without postulating fermionic statistics at fundamental level.
- Tsirelson-level Bell correlations *can* arise from purely constraint-driven finite-Hilbert-space dynamics.

Key distinction:

- **Fundamental fermions**: Particles defined with built-in anticommutation relations
- **Emergent fermionic behavior**: Collective excitations exhibiting antisymmetry and exclusion as derived properties

Our work explores the latter in controlled computational setting.

10.3 Relation to Condensed Matter Phenomena

Similar emergent fermionic behavior appears in:

- Jordan-Wigner transformation of spin-1/2 chains
- Composite fermions in quantum Hall systems
- Parton constructions in spin liquids
- Majorana modes in superconductors

Our work provides different route via constraint-driven dynamics in Hilbert space.

10.4 Methodological Value of Transparent Bug Documentation

We documented bug discovery in unusual detail for research paper. This serves three purposes:

1. **Transparency**: Readers verify core results survive rigorous error-checking
2. **Reproducibility**: Future implementations avoid same pitfalls
3. **Pedagogical**: Debugging process illustrates difference between computational artifacts and genuine physics

The fact that "too perfect" results initially seemed suspicious, prompted validation, revealed actual bugs, yet ultimately proved genuine *strengthens* confidence in findings.

10.5 Scope and Limitations

Clear limitations:

- Lattice sizes are small
- Hamiltonians are hand-crafted
- Models lack relativistic structure
- Gauge symmetry limited to simple $\mathbb{Z}_2$ ingredients
- No derivation from deeper symmetry principles
- No embedding in continuum field theory
- Proton example is purely qualitative

Most importantly: Nothing demonstrates real world operates according to substrate framework. At best, we show such framework is *capable* of reproducing selected features of quantum behavior.

10.6 Future Directions

Immediate next steps:

1. Scale up finite substrates (lattice size, local Hilbert space richness)
2. Use matrix-free and tensor-network techniques
3. Extend to three-body and many-body systems
4. Test disorder and impurity effects

Broader phenomena:

1. Multi-party Bell tests
2. Contextuality scenarios
3. Dynamical processes: scattering, thermalization

4. Connection to gauge theories and Yang-Mills structure

Interpretive:

1. Identify concrete experimental signatures
2. Distinguish constraint-based emergence from conventional unitary-plus-decoherence
3. Analytical investigation of symmetry structure
4. Precise boundaries of robust parameter regions

10.7 Conservative Assessment

Despite limitations, results provide useful milestone:

A simple, finite, constraint-driven substrate can:

- Sit comfortably within quantum correlation polytope
- Reach Tsirelson bound
- Exhibit realistic trade-offs between correlation, entropy, and information flow
- Produce exact antisymmetric-singlet ground states across large parameter regions
- Support both nonlocal quantum correlations and localized composite objects

This gives substrate program firm numerical foothold from which to explore more ambitious connections to real-world physics.

11 Conclusions

We have presented evidence for:

(1) Structurally stable fermionic-like emergence: Two-body substrate Hamiltonians produce ground states with exact antisymmetric exchange symmetry, pure singlet character, and Tsirelson-saturating Bell correlations. These properties persist across $\pm 20\%$ perturbations in six dimensions.

(2) Constraint-driven Bell correlations: Gauss-like consistency penalties, combined with local Hamiltonian dynamics, generate robust CHSH violations reaching Tsirelson bound.

(3) Information-correlation trade-offs: Extracting classical information from Tsirelson-level states requires entropy production and degrades Bell violations in predictable manner.

(4) Methodological rigor: Systematic bug discovery via symmetry-breaking validation demonstrates results survive rigorous scrutiny. Finding and fixing bugs proved robustness of underlying physics rather than undermining it.

These results constitute existence proofs that fermionic-like quantum behavior and Tsirelson-level correlations can emerge from finite-Hilbert-space substrate dynamics without postulating fermionic statistics or additional nonlocal primitives at fundamental level.

While this does not explain real fermions or real quantum mechanics, it demonstrates logically consistent alternatives to standard framework, potentially relevant for:

- Quantum simulation

- Foundations of quantum mechanics
- Emergent phenomena in condensed matter
- Understanding constraints as structural features rather than arbitrary limits

Final principle: Throughout this investigation, we maintained conservative posture—*if the data do not show it, we do not say it.* This discipline, combined with transparent reporting of bugs and validation procedures, provides foundation for future work exploring more ambitious connections to real-world physics.

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References

[1] J. S. Bell, "On the Einstein Podolsky Rosen Paradox," Physics Physique Fizika **1**, 195 (1964).

[2] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, "Proposed experiment to test local hidden-variable theories," Phys. Rev. Lett. **23**, 880 (1969).

[3] B. S. Cirel'son, "Quantum generalizations of Bell's inequality," Lett. Math. Phys. **4**, 93 (1980).

[4] A. Aspect, P. Grangier, and G. Roger, "Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment," Phys. Rev. Lett. **49**, 91 (1982).

[5] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000).

[6] P. Jordan and E. Wigner, "Über das Paulische Äquivalenzverbot," Z. Phys. **47**, 631 (1928).

[7] J. K. Jain, "Composite-Fermion Approach for the Fractional Quantum Hall Effect," Phys. Rev. Lett. **63**, 199 (1989).

[8] N. Read and S. Sachdev, "Large-N Expansion for Frustrated Quantum Antiferromagnets," Phys. Rev. Lett. **66**, 1773 (1991).

[9] A. Y. Kitaev, "Unpaired Majorana Fermions in Quantum Wires," Phys.-Usp. **44**, 131 (2001).